# Practical 1: Introduction to React and Component Architecture

## Practical statement:

• Create a React application using Vite for a student portfolio page.

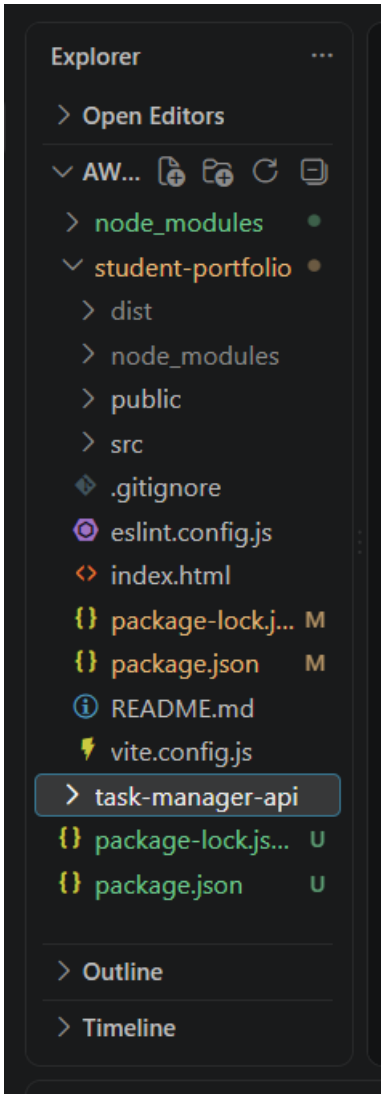
• The application must include at least 4 reusable components: Header, About, Skills, and Footer.

• Each component must be independently structured and composed into a single-page layout.

• No logic or JSX should be duplicated across components.

• Props must be used to pass at least one piece of data into a minimum of 2 components.

## Folder Structure:




## Codes:

## App.jsx

```
import { useState } from "react";
import { Routes, Route } from "react-router-dom";
import "./App.css";

import NavBar from "./components/NavBar";

import Home from "./Pages/Home";
import Project from "./Pages/Projects"; // Make sure the file is Projects.jsx
import Contact from "./Pages/Contact";
import NotFound from "./Pages/NotFound";

function App() {
  const [darkMode, setDarkMode] = useState(false);

  return (
    <div className={darkMode ? "dark app" : "light app"}>
      <NavBar />

      <div className="theme-btn">
        <button onClick={() => setDarkMode(!darkMode)}>
          {darkMode ? "☀ Light Mode" : " ◗ Dark Mode"}
        </button>
      </div>

      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/projects" element={<Project />} />
        <Route path="/contact" element={<Contact />} />
        <Route path="*" element={<NotFound />} />
```

```
      </Routes>
    </div>
  );
}

export default App;
```

## Header.jsx

```
function Header() {
  return (
    <section className="hero">
      <h1>Welcome to My Portfolio</h1>
      <h2>I'm Datri Vasani</h2>
      <p>
        Front-End Developer | React Developer | Web Designer
      </p>

      <button>Explore My Work</button>
    </section>
  );
}

export default Header;
```

## About.jsx

```
function About() {
  return (
    <section className="card">

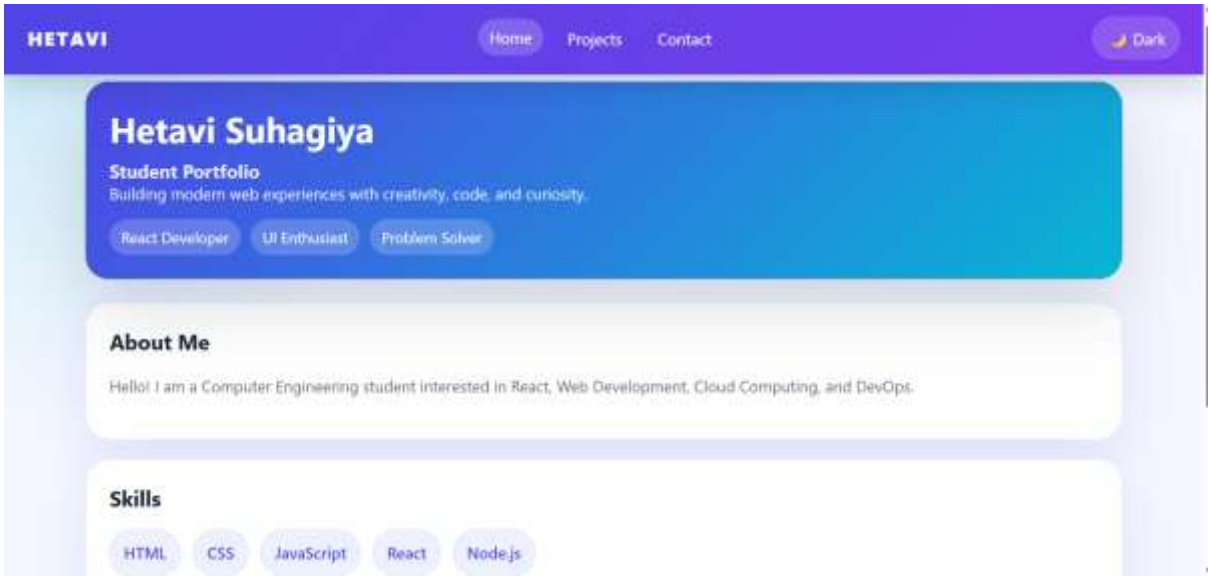
      <h2>About Me</h2>
```

```
      <p>
        Hello! I am Datri Vasani, a passionate Front-End Developer who enjoys
        creating responsive and user-friendly web applications using HTML,
        CSS, JavaScript and React.
      </p>

      <p>
        I enjoy learning new technologies and building modern web interfaces.
      </p>

    </section>
  );
}

export default About;
```




## Skill.jsx

```
function Skill() {
  return (
    <section className="card">
```

```
      <h2>My Skills</h2>

      <div className="skills">

        <div className="skill-box">HTML5</div>

        <div className="skill-box">CSS3</div>

        <div className="skill-box">JavaScript</div>

        <div className="skill-box">React JS</div>

        <div className="skill-box">Bootstrap</div>

        <div className="skill-box">Git & GitHub</div>

      </div>

    </section>
  );
}

export default Skill;
```




## Footer.jsx

```
function Footer() {
```

```
  return (
    <footer className="footer">

      <h3>Datri Vasani</h3>

      <p>
        React Portfolio Project
      </p>

      <p>
        © 2026 All Rights Reserved.
      </p>

    </footer>
  );
}

export default Footer;
```




## Key Question:

**• What is the role of each component in the overall UI structure?**

Each component is responsible for a specific part of the user interface. For example, the Header displays the top section, the Home component displays the main page, the Project component displays project details, and the Footer displays the bottom section. This makes the UI easier to manage and organize.

**• How does React re-render when props change?**

When the props of a component change, React detects the change and re-renders that component. It updates the UI with the new data received from the parent component.

**• Why is component reusability important in large-scale applications?**

Component reusability allows us to use the same component in different parts of an application without writing the same code again. It reduces code duplication, saves time, and makes the application easier to maintain and manage.

## Supplementary Problems:

## NavBar.jsx

```
import { NavLink } from "react-router-dom";


function NavBar() {
  return (
    <nav>


      <NavLink
        to="/"
        className={({ isActive }) =>
          isActive ? "active" : ""
        }
      >
        Home
      </NavLink>


      <NavLink
        to="/projects"
        className={({ isActive }) =>
          isActive ? "active" : ""
        }
      >
        Projects
      </NavLink>
```

```
      <NavLink
        to="/contact"
        className={({ isActive }) =>
          isActive ? "active" : ""
        }
      >
        Contact
      </NavLink>

    </nav>
  );
}

export default NavBar;
```




**GitHub Link:**

https://github.com/hetavisuhagiya/AWDF-24DCE140